

EDRIENE JAY O. CABANELA

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PROFESSIONAL SUMMARY

Computer Engineering graduate specializing in System and Network Administration with hands-on experience in software, web, mobile, and IoT application development through academic projects and industry internship. Experienced with Python, Django, Flask, Laravel, PHP, Kotlin, Java, MySQL, RESTful APIs, Linux, and cloud-based technologies. Familiar with application testing, deployment, database integration, version control, and system implementation. Able to work collaboratively in developing and supporting technology solutions while continuously expanding technical skills.

TECHNICAL SKILLS

- **Programming:** Java, Kotlin, Python, PHP, C++
- **Web Development:** HTML, CSS, Tailwind CSS, Django, Flask, Laravel
- **Database Management & Cloud:** MySQL, Firebase, AWS EC2
- **API & Software Development:** RESTful APIs, JSON, Git, GitHub, Docker
- **Systems & Networking:** Windows & Linux Administration, Network Configuration, Cisco Packet Tracker, NGINX, Apache, Unicorn, VNC, PuTTY, CCTV Installation
- **IoT & Embedded Technologies:** Raspberry Pi, Arduino, ESP8266, ESP32, ARCore, NRF24L01
- **Development Environments:** Android Studio, Visual Studio Code, Python Virtual Environments
- **UI/UX & Graphic Design:** Figma, Canva, Adobe Photoshop, Adobe InDesign
- **Productivity & Collaboration:** Microsoft 365, Google Workspace, AI Productivity Tools

WORK EXPERIENCE

DEPARTMENT OF THE INTERIOR AND LOCAL GOVERNMENT – CAR, Baguio City

June – July 2025

Web Development Intern

- Developed both frontend and backend components of the Recruitment and Placement Portal using Laravel and MySQL, contributing to applicant, administrator, examination, and recruitment management functionalities.
- Developed the Personal Data Sheet (PDS) and Work Experience Sheet (WES) Builder, including the application logic and PDF-ready output mapping based on the required document formats.
- Implemented OTP-based registration and account recovery functionalities and configured SMTP email services to support system communications.
- Contributed to database design by developing table structures, relationships, and Laravel migrations, while working across models, controllers, and views to implement application logic.
- Designed and implemented user interface components for applicant management, examination management, and administrative functions, with emphasis on usability and workflow.
- Conducted module-level and manual testing, identified and reported bugs, and assisted in debugging and resolving issues throughout development.
- Configured and deployed the web application on Hostinger, assisting with production environment setup and deployment.
- Collaborated with the project supervisor and development team to understand requirements, implement features, and incorporate feedback into the system.

PROJECTS & HANDS-ON EXPERIENCES

ARISE: AR-INTEGRATED SPATIAL EXPLORER

May 2026

Capstone Project | Android Developer & Technical Writer | Kotlin, Java, ARCore, Django, MySQL, Raspberry Pi 5

- Developed the Android application, user interface, and core application logic for an augmented reality indoor navigation system providing marker-assisted navigation and contextual information within institutional environments.
- Integrated the native Android application with the Unity-based AR module and Django backend through HTTP requests and APIs, managing the Unity-to-Android Studio integration and client-server communication.
- Assisted in deploying the backend environment on a Raspberry Pi 5 using Docker and Cloudflare Tunnel for centralized data management, remote access, and application synchronization.
- Implemented application workflows for retrieving room, facility, and contextual information managed through the Django-based administrative interface.
- Evaluated system performance across marker detection, navigation accuracy, latency, rendering performance, contextual information delivery, and usability, achieving an 86.67% marker detection rate, 0.3603 m mean navigation error, 255.56 ms average latency, and 29.76 FPS average rendering performance.
- Achieved 100% accuracy in contextual information and notification delivery, with a 4.5200 overall usability weighted mean and 95.56% positive response rate during user evaluation.
- Served as the primary technical writer for the capstone manuscript and prepared supporting technical documentation and project communication materials.

STUDY & QUIZ MANAGEMENT SYSTEM

Web Developer | PHP, Laravel, SQLite, Tailwind CSS

- Contributed to the development of a Laravel-based study and quiz management system for an external project.
- Designed the user interface and contributed to frontend implementation across study, quiz, and administrative modules.
- Assisted with backend development and application logic for managing study materials and quiz questions.
- Contributed to the development of administrative functionality for uploading and managing learning materials and assessment content.
- Assisted in integrating frontend and backend components to support the system's study and assessment workflows.
- Assisted with deployment and hosting of the Laravel application on Render.

DILG-CAR RECRUITMENT AND PLACEMENT PORTAL

July 2025

Web Developer | PHP, Laravel, MySQL, Tailwind CSS

- Developed frontend and backend components of a recruitment and placement portal designed to digitize applicant and recruitment management processes previously handled through manual and online form-based workflows.
- Developed the Personal Data Sheet (PDS) and Work Experience Sheet (WES) Builder, including application logic and data mapping to generate PDF-ready documents based on required formats.
- Implemented OTP-based registration and account recovery with SMTP email integration for automated user verification and account recovery.

- Contributed to applicant, administrator, and examination management functionalities, developing both user interface components and backend application logic.
- Designed database tables, relationships, and Laravel migrations, and worked across models, controllers, and views to implement system functionality.
- Conducted module-level and manual testing, identifying and reporting bugs throughout development and assisting in debugging and refinement.
- Configured the application for deployment on Hostinger, gaining hands-on experience with production web application deployment.
- Improved the overall recruitment workflow by consolidating applicant information and system processes into a centralized web-based platform, making recruitment-related tasks more organized and accessible.

IoT-BASED HAND-GESTURE CONTROLLED ROBOTIC ARM

May 2025

Developer | Python, Flask, Raspberry Pi 4B

- Developed a Python Flask server on Raspberry Pi 4B to receive and process motion data from a smartphone over a local Wi-Fi network.
- Implemented the application logic for processing accelerometer and gyroscope data and translating motion inputs into control commands for the robotic arm.
- Developed the communication and control workflow between the smartphone client, Flask server, and servo-driven robotic arm to enable wireless operation.
- Contributed to the mechanical and system design of the robotic arm and assisted in calibrating motion inputs for reliable operation.
- Conducted accuracy testing and calibration to improve the consistency of gesture-based control.
- Successfully demonstrated wireless, real-time operation and control of the robotic arm through smartphone motion inputs.

HAND-GESTURE CONTROLLED REMOTE CONTROL CAR

April 2025

Developer | Arduino Nano, C++, NRF24L01

- Designed the vehicle chassis and contributed to the development of a hand-gesture-controlled remote control car using Arduino Nano.
- Implemented wireless communication using NRF24L01 modules to transmit gesture-based control commands between the controller and vehicle.
- Developed and integrated the control logic for responsive, real-time vehicle operation based on gesture inputs.
- Successfully demonstrated wireless gesture-based control of the vehicle through the implemented communication and control system.

THE VALLEY DIGITAL: ONLINE NEWS PLATFORM

February 2025

Lead Developer | Python, Django, Django REST Framework, AWS EC2

- Led the development of a full-stack online news platform using React for the frontend and Django/Django REST Framework for the backend.
- Designed and implemented backend models, relationships, CRUD functionality, RESTful APIs, and Django administrative features for managing articles, categories, sections, tags, authors, and media.
- Developed the frontend user interface and integrated React components with backend APIs to support content browsing, search, publishing, and media-driven content workflows.
- Implemented administrative publishing functionality and content management features, including article organization, author management, media uploads, and category/tag management.
- Integrated a Facebook comment plugin and implemented search functionality to enhance reader interaction and content discovery.
- Configured and deployed the application on AWS EC2, handling server setup, web server configuration, SSL, domain configuration, and production deployment.
- Troubleshooted and resolved deployment-related issues involving static and media files to ensure proper asset delivery in the production environment.
- Coordinated development tasks, distributed responsibilities among team members, and integrated individual contributions into the completed system.
- Successfully deployed and turned over the completed platform to the client for use.

WIRELESS WORMS 2.0: IoT-BASED VERMICULTURE MONITORING SYSTEM

May 2024

Developer | Java, Firebase, ESP8266

- Developed the Android application and Firebase backend for an IoT-based vermiculture monitoring and control system.
- Designed and implemented the Android user interface for viewing real-time environmental conditions, including temperature, humidity, and moisture readings.
- Implemented Firebase functionality for real-time data synchronization, user accounts, and storage of historical sensor data.
- Integrated the Android application with an ESP8266-based IoT system that collected sensor readings and controlled connected actuators.
- Implemented remote actuator control and threshold-based notifications to alert users when monitored conditions reached configured limits.
- Conducted manual testing of the Android application, real-time data synchronization, sensor monitoring, and remote control functionality.
- Successfully demonstrated the system's monitoring, notification, and remote-control functions, with the Android component serving as the primary area of contribution.

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900) — April 2026**
Cloud concepts, Azure services, security, and cloud management.
- **Google Cloud Certified: Cloud Digital Leader — March 2026**
Cloud technologies, digital transformation, and business applications.
- **Civil Service Eligibility — Professional Level — March 2026**
- **Basic Occupational Safety and Health (BOSH) — Safety Officer 2 — February 2025**

LEADERSHIP EXPERIENCES

Institute of Computer Engineers of the Philippines Student Edition (ICpEP.SE) – PSU UCC

Technical and Multimedia Director | 2025–2026

Assistant Secretary | 2024–2025

Business Manager | 2023–2024

- Led technical and multimedia activities for the organization, supporting the development and delivery of digital materials and event-related requirements.
- Contributed to the planning and execution of organizational activities, coordinating technical, multimedia, and operational requirements.
- Developed and prepared visual and digital materials for organizational announcements, events, and communications.
- Supported organizational documentation and administrative activities through successive leadership roles.

Institute of Computer Engineers of the Philippines Student Edition (ICpEP.SE) – Region 1

Multimedia Officer | 2025–2026

- Supported regional-level activities through the development and preparation of multimedia and digital communication materials.
- Coordinated multimedia requirements for regional organizational activities and events.

The Technoscope – PSU UCC

Layout Artist | 2024–2026

- Designed and prepared publication layouts and visual materials for organizational and academic content.
- Collaborated with the publication team to develop visually consistent and publication-ready materials.

AWARDS & SCHOLARSHIPS

- **DOST-SEI Junior Level Science Scholarships (JLSS) Merit Scholar** — 2024–Present
Merit-based scholarship awarded under the DOST-SEI Junior Level Science Scholarships program.
- **Department Leadership Award** — ICpEP.SE – PSU UCC | 2026
Recognition for leadership and service within the student organization.
- **Leadership Award for Graduating Class Officers** — Pangasinan State University – Urdaneta City Campus | 2026
Recognition for leadership and service as a graduating class officer.

EDUCATION

PANGASINAN STATE UNIVERSITY – URDANETA CITY CAMPUS

July 2026

Bachelor of Science in Computer Engineering, major in System and Network Administration

Relevant Areas of Study:

- System and Network Administration
- Computer Networks and Network Configuration
- Web and Mobile Application Development
- Database Management
- Internet of Things (IoT) and Embedded Systems
- Software Engineering and Systems Development
- Computer Architecture and Digital Systems

LEARNING CREDENTIALS AND TRAININGS

Google Cloud | March 2026

- **Scaling with Google Cloud Operations** — Google | March 2026
- **Trust and Security with Google Cloud** — Google | March 2026
- **Modernize Infrastructure and Applications with Google Cloud** — Google | March 2026
- **Innovating with Google Cloud Artificial Intelligence** — Google | March 2026
- **Exploring Data Transformation with Google Cloud** — Google | March 2026
- **Digital Transformation with Google Cloud** — Google | February 2026

Microsoft Azure

- **Microsoft Azure Fundamentals: Building Your Cloud Foundation** — MISNET Education Inc. | February 2026
- **Describe Cloud Computing** — Microsoft | April 2025
- **Describe the benefits of using Cloud Services** — Microsoft | April 2025
- **Describe Cloud Service Types** — Microsoft | April 2025
- **Describe Azure Identity, Access, and Security** — Microsoft | February 2026
- **Describe Monitoring Tools in Azure** — Microsoft | February 2026
- **Describe the Core Architectural Components of Azure** — Microsoft | April 2026
- **Describe Azure Compute and Networking Services** — Microsoft | April 2026
- **Describe Azure Networking Services** — Microsoft | April 2026
- **Describe Azure Storage Services** — Microsoft | April 2026
- **Describe Cost Management in Azure** — Microsoft | April 2026
- **Describe Features and Tools in Azure for Governance and Compliance** — Microsoft | April 2026
- **Describe Features and Tools for Managing and Deploying Azure Resources** — Microsoft | April 2026
- **Describe Azure Monitoring Tools** — Microsoft | February 2026

Other Trainings and Seminars

- **Introduction to Cybersecurity** — Cisco Networking Academy | January 2026
- **Work Smarter, Not Harder: Optimize Work with AI** — DICT-CAR | January 2026
- **Seminar on Job Hunting and Exit Conference** — DOST-R1 | August 2026
- **3rd Annual Conference of the Center for Integrated Circuit and Device Research (CIDR)** — CIDR | August 2026
- **Web Content Management Using WordPress** — DICT-CAR | July 2025
- **Web Development Workshop** — Zuitt | July 2024
- **The Future of Development: Will Vibe Coding Replace Traditional Programming?** — ICpEP | June 2025
- **Artificial Intelligence/Machine Learning Webinar** — USAID | October 2024